Python & MATLAB LLT Solver Validation

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Overview

This report documents the validation of both the Python and MATLAB implementations of Prandtl's Lifting-Line Theory (LLT) against classical analytical results. The theory is drawn from Anderson's Anderson's extitFundamentals of Aerodynamics (John D. Anderson, 2011, Chapter 5). The solver is tested on an untwisted elliptic planform of aspect ratio $AR = 8$ using thin-airfoil input data ($c_l = 2\pi\alpha$, $c_d = 0$).

Validated at commit: `e807286` on branch `integrated`.

Results apply to this specific version only; subsequent commits should be re-validated.

Lift Slope

For an elliptic wing in inviscid flow, classical lifting-line theory predicts a three-dimensional lift slope of

$$a = \frac{a_0}{1 + a_0/(\pi AR)}, \tag{1}$$

where $a_0 = 2\pi$ is the thin-airfoil sectional lift slope when sections are assumed to behave according to thin-airfoil theory. Equation (1) is Anderson's Eq. (5.69), on pg. 443, and substitution of the thin-airfoil lift slope yields

$$a = \frac{2\pi AR}{AR + 2}, \tag{2}$$

which recovers the 2D lift slope in the limit of $AR \to \infty$.

For $AR = 8$, equation (1) gives $a = 5.0265$ rad^{-1} (0.08773 deg^{-1}). Both solvers were run at angles of attack from -10° to $+10^\circ$ in 2° increments using 31 spanwise stations. The results are compared in Table eftab:liftslope.

Table 1: Comparison of lift slope from theory, Python, and MATLAB implementations (31 stations).

	Theoretical	Python	MATLAB
$dC_L/d\alpha$ (rad^{-1})	5.0265	4.9957	4.9957
Error (%)	—	0.61	0.61

Both implementations produce identical results to six decimal places, with a relative error of 0.61% at 31 stations. This small discrepancy can be attributed to the discrete approximation (31 spanwise stations, trapezoidal integration), as shown by the decreasing error with increasing N in the convergence study described below.

The lift curve is plotted in Figure 1, where the computed results from both implementations and the theoretical slope are nearly indistinguishable.

Induced Drag

For an elliptic wing, the induced drag coefficient is given exactly by

$$C_{D_i} = \frac{C_L^2}{\pi AR}. \tag{3}$$

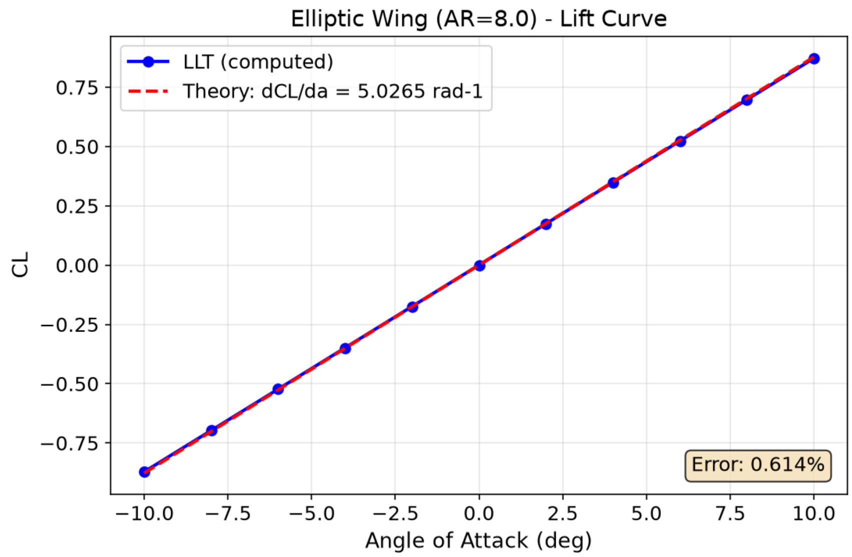

Figure 1: Lift curve for the elliptic wing ($AR = 8$). The computed results (blue markers) and the theoretical slope (red dashed line) are nearly indistinguishable.

This is Anderson's Eq. (5.61) with the induced drag factor $\delta = 0$ for an elliptic planform (John D. Anderson, 2011, pp. 464–465).

The drag polar is shown in Figure 2. Both solvers reproduce the theoretical parabola $C_{D_i} = C_L^2/(\pi AR)$ nearly perfectly, confirming the correct induced-drag calculation.

Convergence Study

A convergence study was performed to assess the effect of spanwise discretisation on the lift-slope error. Both solvers were run at five station counts spanning the range 31 to 601.

Table 2: Convergence of lift slope error with increasing spanwise stations.

Stations	$dC_L/d\alpha$ (rad^{-1})		Error (%)
	Python	MATLAB	
31	4.995672	4.995672	0.614
151	5.018057	5.018057	0.169
301	5.021156	5.021156	0.107
601	5.022749	5.022749	0.076

The convergence behaviour is illustrated in Figure 3.

The Python and MATLAB results are identical at every station count, confirming that both implementations compute the same numerical solution.

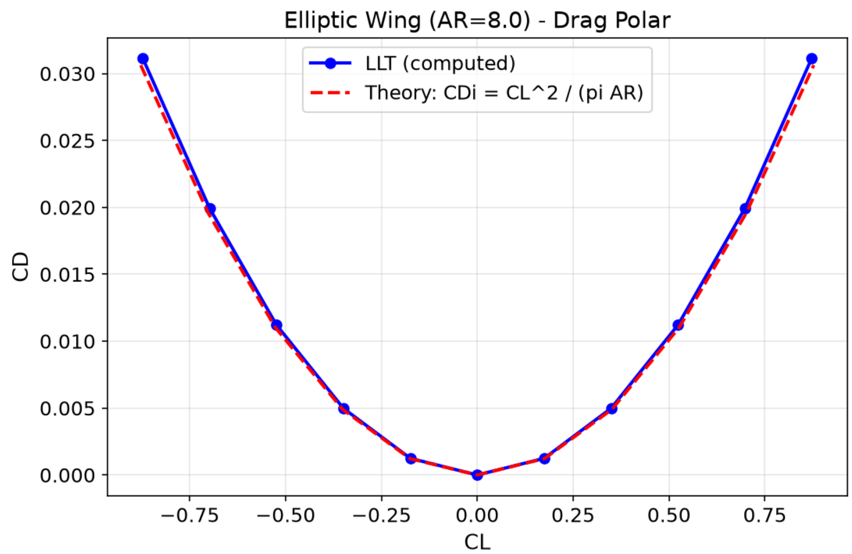

Figure 2: Drag polar for the elliptic wing ($AR = 8$). The computed results follow the theoretical parabola.

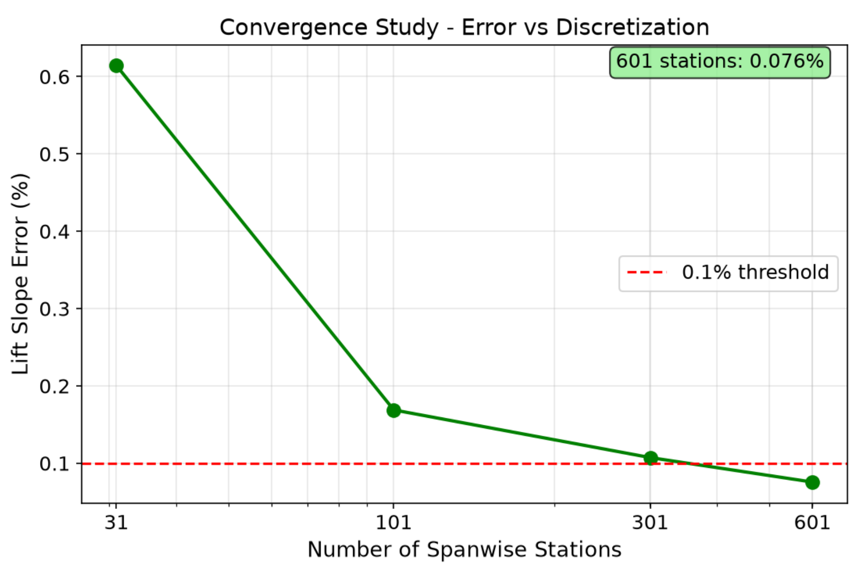

Figure 3: Convergence of the lift-slope error with increasing number of spanwise stations. The 0.1% threshold is reached at approximately 601 stations.

Summary

Both the Python and MATLAB lifting-line solvers have been validated against classical theory John D. Anderson (2011) for an elliptic wing. The key results are:

- Python and MATLAB implementations produce **identical** numerical results (to six decimal places).
- Lift slope error: 0.61% at 31 stations, 0.076% at 601 stations.
- Induced drag follows the theoretical $C_{D_i} = C_L^2/(\pi AR)$ relation.
- Both solvers converge monotonically with increasing discretisation.

Both solvers are ready for use on arbitrary planforms and can accept either thin-airfoil theory or empirical airfoil data.

References

John D. Anderson, J. *Fundamentals of Aerodynamics*. McGraw-Hill, 5th edition, 2011. ISBN 978-0-07-339810-5.